Shafayet Sadik Sowad

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ShafayetSadik | Shafayet Sadik Sowad

Summary

EEE graduate and Systems Engineer specializing in autonomous drones, embedded hardware, and multi-domain system integration. Experienced in applying machine learning to wireless communication and UAV systems. Adept at bridging hardware and software, with a strong research drive and commitment to academic collaboration.

Research Interests

Autonomous UAV Inspection • Path Planning • UAV Photogrammetry & 3D Reconstruction • Machine Learning for Aerial Perception • Digital Twins

Education

Islamic University of Technology (IUT)
B.Sc in Electrical and Electronic Engineering **October 2025**
CGPA: 3.64/4.00
Thesis: *Machine Learning-Driven Handover Optimization in LTE Networks: Quality Signal Indicator, Classifier-Based Prediction Analysis and Hybrid Q-Learning*

– Conducted a drive-test-based study of LTE networks in Dhaka to build a real-world dataset focused on signal quality indicators for handover analysis. Developed machine learning and hybrid Q-learning frameworks to predict and optimize handover decisions.

Publications

Journal Papers

- M. M. S. Shafi, K. M. Istiaque, **S. S. Sowad**, and M. T. Kawser, "Drive-Test-Based LTE Handover Dataset for Cellular Mobility Studies in Urban Bangladesh," *Data in Brief*, 2026, Art. no. 113220, doi: https://doi.org/10.1016/j.dib.2026.113220

Conference Papers

- K. M. Istiaque, **S. S. Sowad**, M. M. S. Shafi, and M. T. Kawser, "Reducing Handover and Ping-Pong Events in LTE Networks via Q-Learning and Subtractive Clustering," *2025 8th International Seminar on Research of Information Technology and Intelligent Systems (ISRITI)*, Yogyakarta, Indonesia, 2025, doi: https://doi.org/10.1109/ISRITI68345.2025.11393278
- K. M. Istiaque, M. M. S. Shafi, **S. S. Sowad**, and M. T. Kawser, "Adaptive Handover Optimization in LTE: Comparative Evaluation of Fuzzy Logic and DQN," *2025 IEEE Asia Pacific Conference on Wireless and Mobile (APWiMob)*, Bali, Indonesia, 2025, pp. 198–204, doi: https://doi.org/10.1109/APWiMob67231.2025.11269172
- M. M. S. Shafi, K. M. Istiaque, **S. S. Sowad**, and M. T. Kawser, "Handover Optimization in LTE Networks Using Contextual Bandit Reinforcement Learning and Real-World Data," *2025 International Conference on Computer Engineering, Network, and Intelligent Multimedia (CENIM)*, Surabaya, Indonesia, 2025, pp. 249–254, doi: https://doi.org/10.1109/CENIM67940.2025.11326341

Publications: Under Review

- M. M. S. Shafi, K. M. Istiaque, T. S. Ahona, **S. S. Sowad** and M. T. Kawser, "Triplet-Loss-Driven Contrastive Autoencoding for Anomaly-Based Handover Prediction Using Drive-Test Data in Dynamic Cellular Environment".
(This journal article has been submitted to *Ad Hoc Networks (Elsevier)* and is currently under review)

Datasets

Drive-Test-Based LTE Handover Dataset for Cellular Mobility Studies in Urban Bangladesh

Open LTE drive-test dataset collected in urban Dhaka, with timestamped RSRP, RSRQ, CINR, velocity, and cell-level measurements in raw and ML-ready formats.

Professional Experience

ANTS Drone Hub — *Systems Engineer* **October 2025 - Present**

– Built a **drone-based window view simulation workflow** for real estate, with **facade viewpoint generation**, a **custom path planner** for drones, and a **Gazebo** simulator.

– Developed software converting **UAV photogrammetry** into OGC 3D Tiles for a Three.js web-based **digital twin viewer**.

- Worked on **aerial thermography** capturing IR images with drones to identify hotspots on solar panels using **ML pipelines**.
- Developed **Swarm drone systems** with synchronized flight control and formation execution, including a **custom light show designer**.
- Planned and led **Agricultural drone spray operations and field testing for precision mapping and autonomous flight missions**.
- Performed **agricultural drone mapping** over approximately 100 sq. km for survey and planning applications.
- Designed circuit schematics and PCBs, including custom library development, filtering, redundancy, and signal integrity practices.

Cybernetics Hi-Tech Solutions (Pvt.) Ltd. — *R&D Engineer Intern* **June 2024 - October 2025**

- **Selected motors and rotary actuators** based on mechanical and electrical requirements, including speed, torque, gearbox selection, and **datasheet analysis.**
- Worked on **drone communication systems**, studying **SBUS, UART, USB, and MAVLink-based data transfer**.
- Contributed to **circuit design and schematic** development.

Skills

Robotics & Autonomy:	ROS2, SLAM, Linux, Kinematics, Gazebo, PX4/ArduPilot, MAVLink, OpenCV.
Programming & ML:	Python, C/C++,MATLAB, Git, TensorFlow, RL, LaTeX.
Embedded & Comms. :	STM32, ESP32,Raspberry Pi,PlatformIO, SBUS, UART, UDP.
Circuit Design & VLSI:	Cadence Virtuoso, Proteus, Simulink, PSpice, Fusion 360, KiCad, Altium .
Software Development:	Three.js, FastAPI, REST APIs, Docker, Webhooks.
Spatial Analysis:	QGIS, Agisoft Metashape.

Projects

MARS Rover

- Developed electrical systems and custom PCBs, integrating autonomous navigation, communication systems.

Autonomous SLAM Drone

- Using Pixhawk, Raspberry Pi, Intel Realsense D435, Optical Flow Sensor.

Underwater ROV Design

- Designed and integrated sensor, power, and communication systems, supporting reliable control and data transmission.

Real-Time IoT based Monitoring, Control and Automated System

- Integrating pH, ultrasonic & temperature sensor, OLED, SIM800L, DC motor, temperature control, and servo motor.

Lipo 6s Battery Management System (BMS)

- A 6s BMS using analog components, including logic gates and op-amps, with Simulink simulation and a functional PCB.

SAP-I 8-bit Computer with ICs, Registers, and Flip-flops

- Built basic computer with hardware components using Proteus for simulation and custom PCBs.

8051 Microcontroller Assembly Coding

- Alarm Clock with real-time countdown, keypad interface, LCD, 7 segment displays and puzzle-solving functionality.

Simulink Model for a Prosthetic Leg Gait Cycle

- Developed a human leg gait cycle model using neural networks and RBF function in SIMULINK.

English Proficiency

IELTS Academic (British Council) **Overall: 7.5/9.0**

- Reading: 7.0 Listening: 7.5 Speaking: 8.0 Writing: 6.5

Extra Curricular Activities

Team Lead	Project Aero (*Drone Project*)	*June 2024 - October 2025*
Vice Chair	IEEE RAS IUT Student Branch Chapter	*June 2024 - October 2025*
Head of Operations	IUT Robotics Society	*August 2024 - October 2025*
Senior Electrical Executive	Project Altair (*Mars Rover*)	*June 2023 - October 2025*
Head of PCB	Accelerate (*Inter University PCB Design Competition*)	*July 2024*

Certificates & Achievements

- **18th Place as Project Altair** — University Rover Challenge 2025, USA
- **2nd Place as Project Aqua** — Underwater Vehicle Design Challenge 2023, IIT Guwahati, India
- **Top 12 Finalist as Project Aero** — International Space Drone Challenge 2024, India
- **6th Place as Project Altair (ABEX Task Winner)** — International Rover Challenge 2024, Coimbatore, India

Reference

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